



Industrial Internship Report

Prilixor Technologies

Submitted by:

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12202040701092

In partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Engineering

Madhuben & Bhanubhai Patel Institute of Technology

The Charutar Vidya Mandal (CVM) University,

Vallabh Vidyanagar – 388120

April - 2026



Madhuben & Bhanubhai Patel Institute of Technology
Computer Engineering

CERTIFICATE

This is to certify that **Manan Donga (12202040701092)** has submitted the Industrial Internship report based on internship undergone at **Prilixor Technologies** for a period of **16 weeks** from **22nd December 2025** to **13th April 2026** in partial fulfillment for the degree of Bachelor of Technology in **Computer Engineering, Madhuben & Bhanubhai Patel Institute of Technology** at The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar during the academic year 2025 – 26.

Mr. Mihir Rajyaguru
Internal Guide

Dr. Gopi Bhatt
Head of the Department



Date: 13/04/2026

TO WHOM IT MAY CONCERN

This is to certify that **Manan Donga**, a student of **Madhuben & Bhanubhai Patel Institute of Technology** has successfully completed his internship in the field of **Software Development** from **22nd December 2025** to **13th April 2026** (Total number of Weeks:**16**) under the guidance of **Mr. Parth Talaviya**.

His internship activities include:

Developed and maintained web applications using Angular and .NET,
Participated in deployment processes.

During the period of her/ his internship program with us, he / she had been exposed to different processes and was found diligent, hardworking, and inquisitive.

We wish him every success in his life and career.

For **Prilixor Technologies**

Authorized Signature with Industry Stamp



DECLARATION

I, **Manan Donga (12202040701092)**, hereby declare that the Industrial Internship report submitted in partial fulfillment for the degree of Bachelor of Technology in Computer Engineering, **Madhuben & Bhanubhai Patel Institute of Technology**, The Charutar Vidya Mandal (CVM) University, Vallabh Vidyanagar, is a Bonafide record of work carried out by me at Prilixor Technologies under the supervision of **Mr. Parth Talaviya** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Prilixor Technologies** for providing me with the opportunity to undertake my internship within the organization. I am especially thankful to all the team members at **Prilixor Technologies** for their support, patience, and the welcoming environment they created, which made this internship a truly enjoyable and enriching experience.

With immense pleasure, I present this internship report as a part of my academic curriculum. I am deeply grateful to everyone who supported and encouraged me from the beginning of this journey.

I would like to extend my heartfelt thanks to my **Faculty Mentor, Internship Coordinator, and Head of Department, Dr. Gopi Bhatt**, for their invaluable guidance and constant encouragement. I also express my appreciation to my **internship guides**, who were always willing to assist me with diverse challenges throughout the internship.

Lastly, I sincerely thank all the faculty, staff members, and colleagues who supported me directly or indirectly in completing this internship successfully.

ABSTRACT

This report presents a comprehensive overview of the development of a full-stack e-commerce web application during my Industrial Internship project. The primary objective was to gain practical experience in modern web development technologies and create a functional online shopping platform.

The project involved developing both frontend and backend components using cutting-edge technologies. On the frontend, I utilized Angular with component-based architecture, services, and routing to create a sophisticated user interface. The application features a professional design with comprehensive e-commerce functionality including advanced shopping cart, authentication system.

On the backend, I implemented a comprehensive RESTful API using ASP .Net Core, with advanced authentication (JWT) and SQL Server. The backend supports complex operations including user management, real-time statistics, product snapshots, and secure file uploads.

Key features of the application includes:

- Dynamic product catalog with filtering, sorting using API integration
- Shopping cart system with persistent storage using local storage and backend sync
- Secure authentication and authorization using JWT tokens
- Order management system with checkout and payment integration via Stripe
- Responsive UI built with Angular and Bootstrap
- Backend developed using ASP.NET Core with clean architecture principles
- Database integration using Entity Framework Core with optimized queries
- Caching implementation using Redis for performance improvement
- RESTful API design with proper error handling and validation
- User order history and detailed product view system

This project significantly enhanced my technical competencies in full-stack web development, particularly in building scalable e-commerce applications. It deepened my understanding of modern frameworks like Angular and ASP.NET Core, database management, API development, and payment integration. It also provided valuable exposure to real-world software architecture and best practices in developing production-ready applications.

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CHAPTER 1: INTRODUCTION

Industrial Internship projects serve as a vital component of engineering education, enabling students to apply theoretical knowledge to real-world applications. They bridge the gap between academic learning and industry practices, helping students gain exposure to modern technologies, frameworks, and software development methodologies.

This report presents the details of my internship project focused on developing a scalable e-commerce web application. The project was carried out as part of the Bachelor of Engineering in Computer Engineering at Madhuben & Bhanubhai Patel Institute of Technology.

During the internship, I worked on a full-stack web development project titled **Skinet**, which is a modern e-commerce platform designed to provide users with a seamless and efficient online shopping experience. The application allows users to browse products, filter and sort items, manage shopping carts, and place orders through a secure checkout process.

My role involved both frontend and backend development. The frontend was developed using Angular to create a responsive and user-friendly interface, while the backend was built using ASP.NET Core to develop robust and scalable APIs. I also worked with Entity Framework Core for database operations and Redis for caching to improve performance.

The application features a modern, responsive design with comprehensive e-commerce functionalities including:

- Home page with product listings and dynamic data loading
- Product catalog with filtering, sorting, and pagination
- Individual product detail pages with dynamic routing
- Shopping cart system with persistent storage
- Secure user authentication and authorization using JWT
- Order management system with checkout and payment integration via Stripe
- RESTful API architecture with proper error handling and validation
- Performance optimization using caching mechanisms
- Responsive UI ensuring compatibility across devices

This report outlines the technologies used, system architecture, development process, challenges encountered, and key learnings achieved during the internship. The experience significantly enhanced my understanding of full-stack development, modern web technologies, and scalable application design, while also improving my problem-solving and practical implementation skills in a professional development environment.

CHAPTER 2: COMPANY PROFILE

2.1 Company Overview

The internship was conducted at Prilixor Technologies, a technology consulting and custom software development company established in 2025. The company specializes in delivering scalable and secure digital solutions for B2B organizations.

Prilixor Technologies focuses on developing web applications, backend systems, and process automation platforms using modern technologies. It supports startups and enterprises by providing end-to-end development services, including design, development, deployment, and maintenance.

During my internship, I worked with the development team and gained practical experience in full-stack development and real-world project workflows. The company follows an agile methodology, ensuring efficient delivery of high-quality and reliable software solutions.

2.2 Technology Infrastructure

Prilixor Technologies maintains a robust and modern technology infrastructure designed to support scalable web application development and enterprise solutions. The company utilizes a wide range of tools and technologies to ensure efficient development, deployment, and maintenance processes.

Development Environment:

The primary development environment is centered around the Microsoft technology stack, including ASP.NET Core and .NET-based architectures for building secure and high-performance backend systems. For frontend development, modern frameworks such as Angular and React are used to create responsive and user-friendly interfaces.

The company also utilizes databases such as Microsoft SQL Server, PostgreSQL, and MongoDB depending on project requirements. Cloud platforms like Microsoft Azure and AWS are leveraged for deployment, scalability, and performance optimization.

Software Development and DevOps:

The organization follows modern development practices, including API-driven architecture and microservices where applicable. Version control systems such as Git are used for code management, along with CI/CD pipelines to automate build, testing, and deployment processes.

Version Control and Collaboration:

Development workflows utilize standard version control systems and collaborative tools to manage projects effectively. They follow a structured software development lifecycle that includes requirement analysis, prototype design, development, quality assurance, delivery, and maintenance.

CHAPTER 3: TECHNOLOGY LEARNED

During my internship, I worked with modern full-stack web development technologies while developing the **Skinet** application. This experience enhanced my understanding of scalable application development, API integration, and real-world software architecture. The technologies learned are outlined below:

3.1 Frontend Technologies

3.1.1 Angular

Angular was used as the primary frontend framework. I learned how to:

- Develop single-page applications using component-based architecture
- Use Angular modules, components, and services
- Implement data binding and dependency injection
- Handle HTTP requests and API integration
- Create reusable UI components
- Implement routing and navigation using Angular Router
- Manage application state effectively

3.1.2 TypeScript

- Strongly typed JavaScript for better code quality
- Use of interfaces, classes, and decorators
- Improved error handling and maintainability
- Modular code structure

3.1.3 HTML5 & CSS3

- Semantic HTML for structured web pages
- Responsive design using Flexbox and Grid
- Styling using modern CSS techniques
- Integration with Bootstrap for UI design

3.1.4 UI & Routing

- Dynamic routing using Angular Router
- Responsive layouts for multiple devices
- Clean UI design with reusable components
- User-friendly navigation and product display

3.2 Backend Technologies:

3.2.1 ASP.NET Core

ASP.NET Core was used for backend development. I learned:

- Building RESTful APIs
- Implementing controllers and services
- Middleware configuration and request pipeline
- Dependency Injection for scalable architecture
- Handling HTTP requests and responses

3.2.2 API Development

- Designing RESTful endpoints
- Handling CRUD operations
- Implementing proper status codes and responses
- Error handling and validation

3.2.3 Authentication & Security

- JWT-based authentication
- Secure API endpoints
- Role-based authorization
- Password security and validation

3.3 Database & Data Management

3.3.1 Entity Framework Core

Entity Framework Core was used for database operations:

- ORM concepts and database mapping
- Writing LINQ queries
- Managing database migrations
- Performing CRUD operations

3.3.2 Database Systems

- Working with relational databases like Microsoft SQL Server
- Data modelling and relationships
- Efficient querying and data handling

3.3.3 Caching with Redis

Redis was used to:

- Improve application performance
- Store frequently accessed data
- Reduce database load

3.4 Payment Integration

3.4.1 Stripe Integration

- Payment processing using Stripe
- Secure checkout implementation
- Handling payment intents and confirmations

3.5 Development Tools

3.5.1 Visual Studio & VS Code

- Backend development using Visual Studio
- Frontend development using VS Code
- Debugging and code management

3.5.2 Version Control

- Git for version control
- Structured project workflow

By the end of the internship, I developed a strong understanding of how frontend, backend, database, and cloud technologies integrate to build a complete web application. I also gained practical experience in developing scalable e-commerce systems, following industry best practices, and working in a professional development environment.

CHAPTER 4: OUTLINE OF WORK DONE DURING INTERNSHIP

The internship work was focused on developing a full-stack e-commerce web application based on **Skinet**. The development process was divided into multiple phases, covering backend development, frontend implementation, and integration of advanced features.

Below is the detailed outline of the work performed during the internship:

Phase 1: Project Setup and Backend Development

- Analyzed requirements of an e-commerce platform
- Set up backend project using ASP.NET Core
- Designed project architecture following clean architecture principles
- Created RESTful APIs for products, orders, and user management
- Implemented database using Entity Framework Core
- Configured database relationships, migrations, and data seeding
- Implemented filtering, sorting, and pagination for product APIs
- Added caching using Redis for performance optimization

Phase 2: Frontend Development

- Set up frontend project using Angular
- Developed reusable components for UI design
- Implemented product listing and product detail pages
- Integrated frontend with backend APIs
- Implemented routing and navigation using Angular Router
- Designed responsive UI using Bootstrap
- Created user-friendly interface for browsing products

Phase 3: Core Features Implementation

- Developed shopping cart functionality with persistent storage
- Implemented user authentication and authorization using JWT
- Created order management system with checkout functionality
- Integrated payment system using Stripe
- Implemented user order history and order tracking
- Added form validation and error handling mechanisms

Phase 4: Integration, Testing, and Optimization

- Integrated frontend and backend modules
- Performed testing of APIs and UI components
- Optimized application performance using caching and efficient queries
- Improved user experience with responsive design and smooth navigation
- Fixed bugs and ensured application stability
- Prepared the application for deployment

Key Achievements

1. Developed a complete full-stack e-commerce application
2. Gained hands-on experience with Angular and ASP.NET Core
3. Implemented secure authentication and payment integration
4. Built scalable APIs with proper architecture
5. Improved performance using caching techniques
6. Created a responsive and user-friendly interface

CHAPTER 5: IMPLEMENTATION AND RESULTS

This chapter presents the detailed implementation of the Skinet e-commerce web application, showcasing the key features, user interfaces, and functionality developed during the internship project.

5.1 Application Architecture

The application follows a modern full-stack architecture with clear separation of frontend and backend:

- **Frontend:** Angular
- **Backend:** ASP.NET Core
- **Database:** Microsoft SQL Server
- **ORM:** Entity Framework Core
- **Caching:** Redis
- **Styling:** Bootstrap

5.2 Homepage Implementation

The homepage serves as the main entry point of the application.

Key Features:

- Product listing with dynamic data from APIs
- Clean and responsive navigation header
- Featured products display
- Responsive design for multiple devices

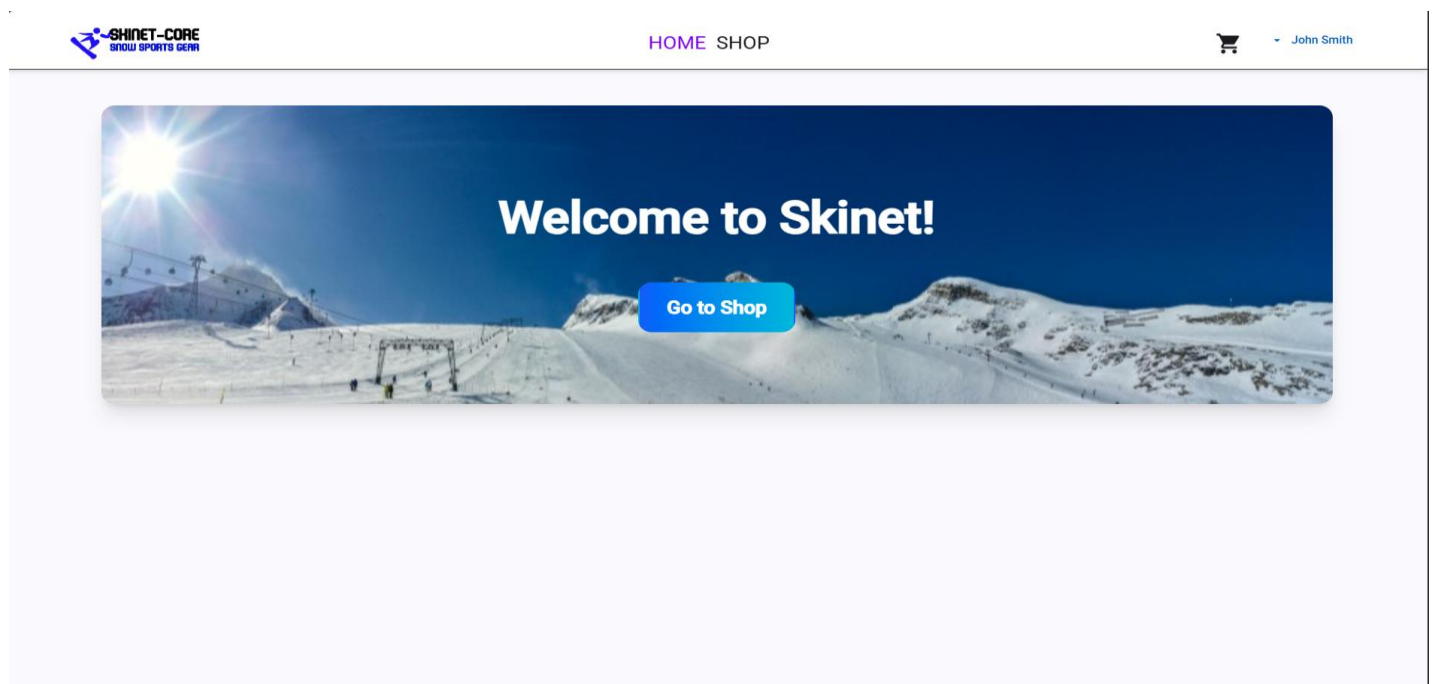


Fig 5.1 Home Page

5.3 Product Catalog Page

The shop page displays all products with advanced filtering options.

Key Features:

- Product listing with pagination
- Filtering by category and sorting options
- Product cards with image, price, and details
- API integration for real-time data

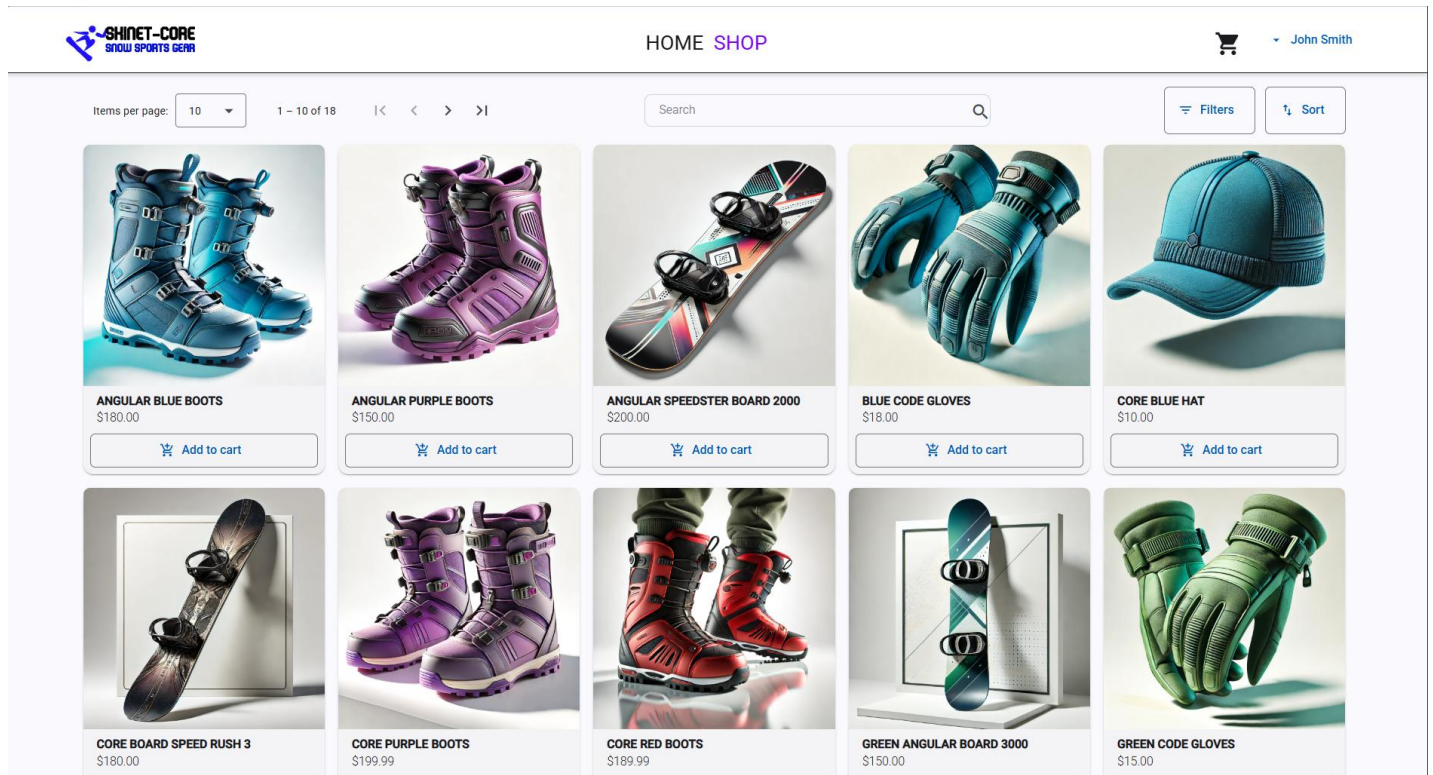


Fig 5.2 Shop Page

5.4 Product Detail Page

Individual product pages provide detailed information about specific products with enhanced user interaction capabilities.

Key Features:

- Dynamic routing based on product ID
- Product description, price, and image
- Add to cart functionality
- Clean UI for better user experience

**Typescript Entry Board**

You have 0 of this item in your cart

\$120.00 Add to Cart

Quantity

1

Aenean nec lorem. In porttitor. Donec laoreet nonummy augue.

Fig 5.3 Product Detail Page

5.5 Shopping Cart System

Angular service-based cart management with backend database persistence for professional e-commerce functionality.

Frontend Cart Features:

- Display of selected products with image, name, and price
- Quantity controls using increment (+) and decrement (−) buttons
- Option to remove items directly from the cart
- Real-time update of cart values without page reload
- Order summary panel showing subtotal, discount, delivery fee, and total
- Voucher code system for applying discounts
- Checkout and continue shopping options for better navigation

Backend Cart Persistence:

- Cart data stored and managed through secure APIs
- User-specific cart association for logged-in users
- Real-time synchronization between frontend and backend
- Product validation and price consistency checks
- Support for discount and pricing calculations at backend level
- Optimized performance using Redis caching



Angular Speedster Board 2000

 Delete

— 1 +

\$200.00



Blue Code Gloves

 Delete

— 3 +

\$18.00

Order Summary

Subtotal	\$254.00
Discount	\$25.40
Delivery Fee	\$0.00
Total	\$228.60

Checkout

[Continue Shopping](#)

Do you have a voucher code?

10% Discount applied

Voucher code

Gimme10

Apply code

Fig 5.4 Shopping Cart Page

5.6 User Authentication System

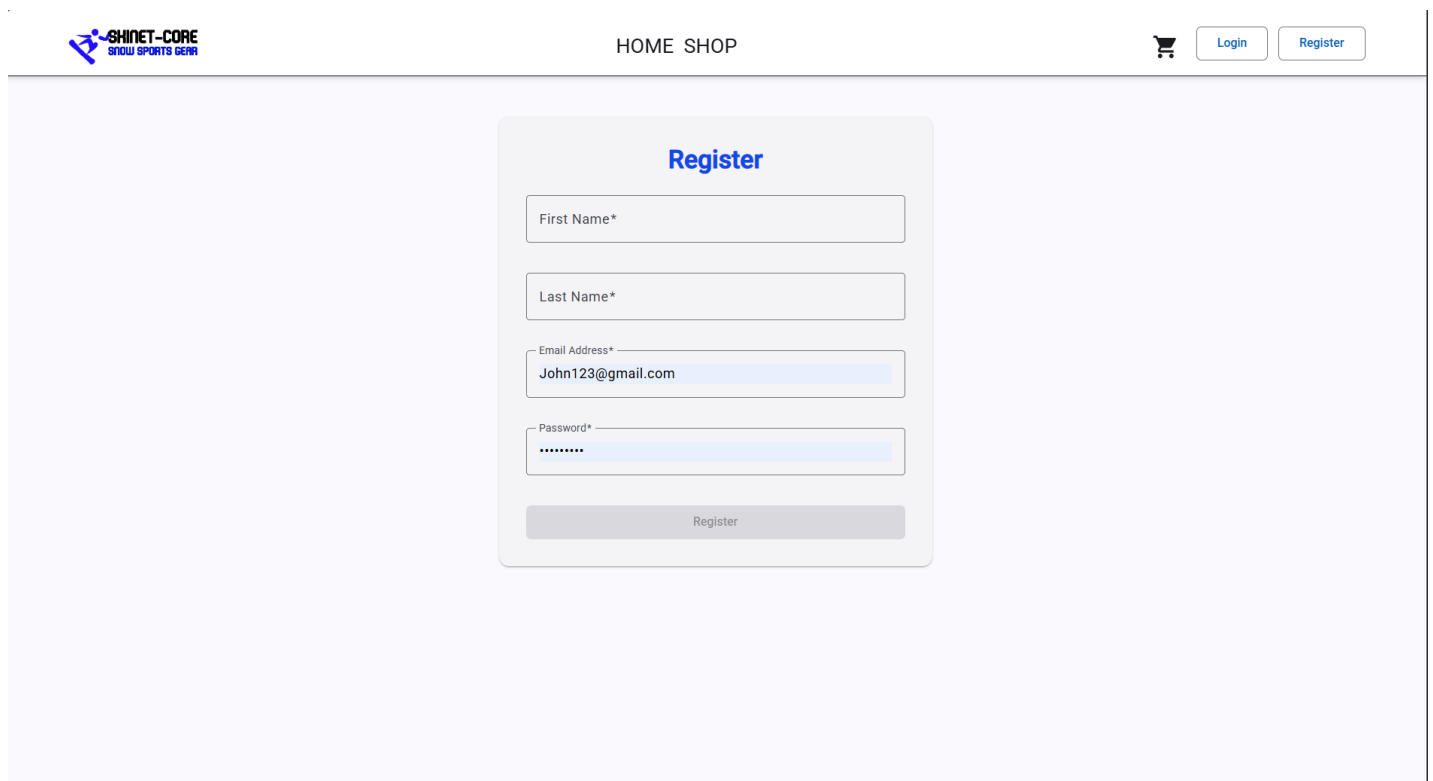
The authentication system provides secure user login and registration functionality.

Login Page Features:

- Email and password input fields
- Form validation and error handling
- Secure authentication using JWT tokens
- Integration with authentication service
- Responsive UI with Bootstrap styling

Registration Page Features:

- User registration form with required details
- Password and confirm password validation
- Input validation and error messages
- Automatic login after successful registration
- Secure data handling and API integration



The screenshot displays a web application's registration page. At the top left is the 'SHINET-CORE SNOW SPORTS GEAR' logo. The top navigation bar includes 'HOME' and 'SHOP' links, a shopping cart icon, and 'Login' and 'Register' buttons. The main content area features a central 'Register' form with the following fields: 'First Name*', 'Last Name*', 'Email Address*' (containing 'John123@gmail.com'), and 'Password*' (masked with dots). A 'Register' button is positioned at the bottom of the form.

Fig 5.5 Registration Page

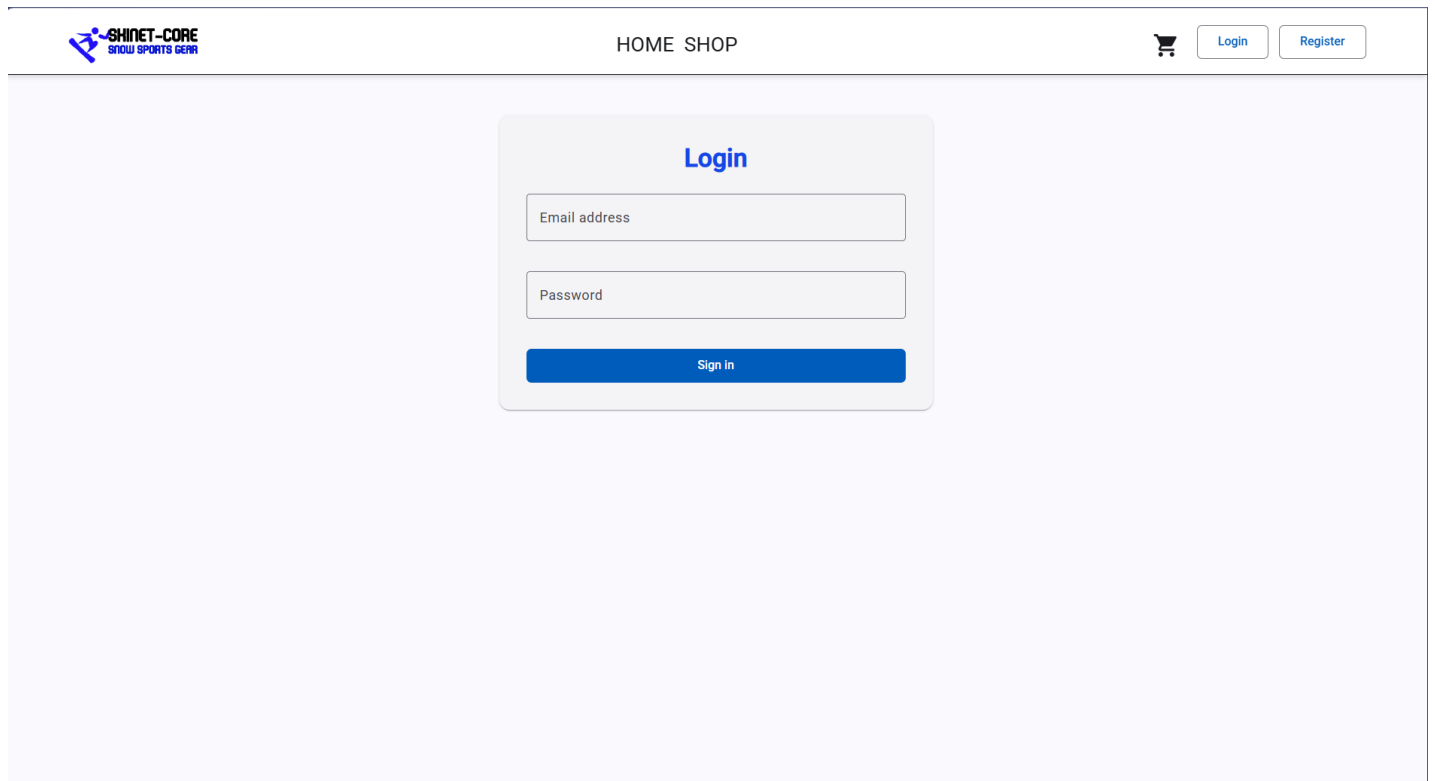


Fig 5.6 Login Page

5.7 Backend API Implementation

The backend provides secure and scalable APIs.

Key Features:

- RESTful API design
- CRUD operations for products and orders
- Error handling and validation
- Secure endpoints with authentication

5.8 Responsive Design Implementation

The application is fully responsive across devices.

Key Features:

- Mobile-first design approach
- Responsive layouts using Bootstrap
- Optimized UI for different screen sizes

5.9 Component Architecture

The application follows a modular component architecture.

Reusable Components:

- Header and navigation
- Product cards
- Cart components
- Forms and UI elements

5.10 State Management

Efficient state management is implemented using Angular services.

Features:

- Centralized state handling
- API data management
- Cart and user session management

5.11 Error Handling and User Experience

Implementation Features:

- Proper error handling for API calls
- User-friendly messages
- Form validation
- Smooth user interaction

5.12 Performance Optimization

Key Optimizations:

- Caching using Redis
- Optimized API calls
- Efficient database queries
- Faster page loading

5.13 Results and Achievements

The developed application successfully demonstrates:

1. Full-stack e-commerce application development
2. Integration of frontend and backend systems
3. Secure authentication and payment system
4. Scalable and maintainable architecture
5. Responsive and user-friendly design
6. Real-world implementation of modern technologies

The project successfully achieves all initial objectives and provides a solid foundation for a commercial e-commerce platform.

CHAPTER 6: CONCLUSION

The internship project focused on developing a full-stack e-commerce web application based on **Skynet** has been a highly valuable and transformative learning experience. As a Computer Engineering student, this project enabled me to apply theoretical concepts to practical implementation and gain hands-on exposure to modern web development technologies and industry practices.

During this project, I successfully developed a scalable e-commerce platform using Angular for the frontend and ASP.NET Core for the backend. I also worked with Entity Framework Core for database operations and Redis for caching, which helped improve application performance and efficiency.

Key technical skills acquired include:

- Secure authentication system using JWT
- Shopping cart and order management system
- Payment integration using Stripe
- RESTful API development with proper validation and error handling
- Database design and management using Entity Framework Core
- Full-stack development with seamless frontend-backend integration
- Responsive UI design using Bootstrap
- Performance optimization using caching techniques
- Clean and scalable project architecture

This project significantly enhanced my understanding of full-stack development, API integration, and real-world application architecture. Building a complete e-commerce system from scratch helped me improve my problem-solving skills, coding practices, and ability to work with modern frameworks.

Summary

The project successfully achieved its objectives by delivering a fully functional e-commerce application with essential features such as product browsing, cart management, user authentication, order processing, and payment integration. It demonstrates the effective use of modern technologies to build scalable and user-friendly web applications.

Future Enhancements

- Implementation of Admin panel
- Integration of additional payment methods
- Enhancement of UI/UX for improved user experience
- Deployment on cloud platforms for scalability and accessibility

In conclusion, this internship project has strengthened my technical foundation and provided practical exposure to industry-level development practices. The experience gained will be highly beneficial for my future career in software development and will serve as a strong base for building more advanced applications.